Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 4 Lab Guide: Building and Hardening a 3-Tier STP Topology

Field	Detail
Based on	Week 4, STP-Based Layer 2 Attacks lecture (STP fundamentals, path cost and port selection, algorithm, versions, attacks, and six mitigations including BPDU Filter)
Runs in	Week 5 lab session (labs trail lectures by one week)
Learning Outcomes	LO2 (evaluate mitigation techniques) and LO5 (configure security technologies)
Environment	Cisco Packet Tracer
Structure	Part 1: build the topology and test its default behaviour. Part 2: harden it with STP security features and test again.
Submission	Completed answer sheet + screenshots of the verification commands. Submission method per course page

This lab has you build a realistic 3-tier hierarchical network, access, distribution, and core, with deliberate redundant links, then observe how STP behaves on it by default. You'll then apply the hardening features from this week's lecture and confirm both that the network still works and that the hardening actually does something measurable.

A note on Packet Tracer's STP support

Packet Tracer implements Cisco's PVST+ and Rapid PVST+ modes (spanning-tree mode pvst / rapid-pvst). It does not support a standalone IEEE RSTP mode or MSTP (spanning-tree mode mst is not available). This lab uses PVST+ as the default/baseline mode and Rapid PVST+ as the faster-converging mode, which is exactly the comparison the lecture's 'STP vs RSTP' material maps onto. MSTP configuration syntax was covered in lecture for your understanding, but is not something you can configure hands-on in this environment.

Part 1, Build the Topology and Test Default Behaviour

Topology

Device	Layer	Role
C1	Core	Top of the hierarchy. Will be configured as the STP root bridge
D1, D2	Distribution	Each connects to C1 and to each other (backbone link), and to both access switches
A1, A2	Access	Each dual-homed to both D1 and D2 for redundancy; PCs connect here
PC1, PC2	End hosts	PC1 on A1, PC2 on A2. Used to test end-to-end connectivity and convergence

Physical layout: C1 connects to both D1 and D2. D1 and D2 also connect directly to each other (the backbone link) and each connect to both A1 and A2. PC1 sits on A1, PC2 sits on A2. Build this in Packet Tracer before configuring anything, it's the same shape used in this week's lecture diagram.

Port Plan

Switch	Port	Connects to	Mode
C1	Gi0/1	D1	Trunk
C1	Gi0/2	D2	Trunk
D1	Gi0/1	C1	Trunk
D1	Gi0/2	D2 (backbone)	Trunk
D1	Fa0/1	A1	Trunk
D1	Fa0/2	A2	Trunk
D2	Gi0/1	C1	Trunk
D2	Gi0/2	D1 (backbone)	Trunk
D2	Fa0/1	A1	Trunk
D2	Fa0/2	A2	Trunk
A1	Gi0/1	D1	Trunk
A1	Gi0/2	D2	Trunk
A1	Fa0/1	PC1	Access
A2	Gi0/1	D1	Trunk

A2	Gi0/2	D2	Trunk
A2	Fa0/1	PC2	Access

Step 1.1, Configure Trunk and Access Ports

On every switch, configure the inter-switch links as trunks and the PC-facing links as access ports, per the port plan above. Example for C1:

```
enable
configure terminal
interface range gigabitEthernet 0/1 - 2
 switchport mode trunk
 exit
```

Example for A1 (trunk uplinks, access port to PC1):

```
interface range gigabitEthernet 0/1 - 2
 switchport mode trunk
 exit
interface fastEthernet 0/1
 switchport mode access
 exit
```

Repeat the equivalent commands on D1, D2, and A2 using their own port assignments from the plan above.

Step 1.2, Address the PCs

Assign PC1 and PC2 static IP addresses on the same subnet (e.g. 192.168.1.10/24 and 192.168.1.20/24). There's only one VLAN in this topology, so no routing is needed yet.

Step 2, Observe Default STP Behaviour

Before changing anything, see what STP does on its own.

```
show spanning-tree vlan 1
```

Run this on all five switches and answer:

- Which switch is currently the root bridge? Is this the switch you'd have chosen deliberately, or did it happen by default MAC-address tie-break?
- Which port(s) on the access and distribution switches are in a Blocking/Discarding state? These are the redundant links STP has deactivated to prevent a loop.
- Look at the Cost column in D1 and D2's output for the backbone link between them (D1's Gi0/2 and D2's Gi0/2). Using the path-cost logic from this week's lecture, explain which of those two ports became Designated on that segment, and why, referencing the actual cost values you see.

Screenshot show spanning-tree vlan 1 output from at least two switches, with the root bridge and one blocked port clearly visible.

Step 3, Measure Baseline (PVST) Convergence Time

- Start a continuous ping from PC1 to PC2 (ping 192.168.1.20 -t on a Windows-style PC, or repeat pings manually).
- While the ping is running, disconnect one of A1's uplinks (e.g. the D1-A1 link) to force STP to recalculate.
- Count how many pings fail before connectivity resumes. Each failed ping is roughly one second, giving you a rough convergence time.
- Reconnect the link and let the topology stabilise before moving on.

Classic STP/PVST+ convergence is slow. Expect somewhere in the region of 30 to 50 seconds of lost connectivity. Record your measured value.

Record: baseline (PVST+) convergence time = _____ seconds.

Step 4, Switch to Rapid PVST+ and Re-Test

On every switch:

```
configure terminal
spanning-tree mode rapid-pvst
end
```

Repeat the exact same convergence test from Step 3 (disconnect the same link, count failed pings, reconnect).

Record: Rapid PVST+ convergence time = _____ seconds. Compare it to your Step 3 result. You should see a dramatic difference.

Part 2, Configure STP Hardening and Test Again

Apply the following controls on top of the network you just built and tested. Stay in Rapid PVST+ mode from Part 1. Don't revert to PVST+.

Step 2.1, Set a Deterministic Root Bridge

On C1, force it to be the root regardless of MAC address:

```
configure terminal
spanning-tree vlan 1 priority 4096
end
```

Step 2.2, Root Guard

On D1 and D2, apply Root Guard to the ports facing the access switches (their Fa0/1 and Fa0/2):

```
configure terminal
interface range fastEthernet 0/1 - 2
 spanning-tree guard root
 exit
```

Step 2.3, BPDU Guard (on A1 only)

On A1 only, apply PortFast + BPDU Guard to the PC-facing port (Fa0/1):

```
configure terminal
interface fastEthernet 0/1
 spanning-tree portfast
 spanning-tree bpduguard enable
 exit
```

Step 2.4, BPDU Filter (on A2 instead)

On A2, apply PortFast + BPDU Filter to the PC-facing port (Fa0/1), as the alternative to BPDU Guard covered in lecture:

```
configure terminal
interface fastEthernet 0/1
 spanning-tree portfast
 spanning-tree bpdufilter enable
 exit
```

A1 and A2 are now protected differently on purpose. Step 5 has you compare what actually happens on each when a rogue device shows up.

Step 2.5, Loop Guard

On A1 and A2, apply Loop Guard to both uplinks (Gi0/1 and Gi0/2):

```
configure terminal
interface range gigabitEthernet 0/1 - 2
```

```
spanning-tree guard loop
exit
```

Step 2.6, UplinkFast and BackboneFast

On A1 and A2 (access layer), enable UplinkFast:

```
configure terminal
spanning-tree uplinkfast
end
```

On D1 and D2 (distribution layer), enable BackboneFast:

```
configure terminal
spanning-tree backbonefast
end
```

Both features were originally designed to speed up classic 802.1D STP. Since you're already running Rapid PVST+, which has fast convergence built in, don't expect a dramatic further improvement. You're configuring them here for completeness and so you can confirm the commands and their status in show output, not because they'll transform an already-fast network.

Step 3, Verify the Configuration

```
show spanning-tree summary
```

Confirm the output shows Loop Guard, Root Guard (where applied), UplinkFast, and BackboneFast as active/enabled, and that C1 is confirmed as the root bridge.

Screenshot show spanning-tree summary from C1, one distribution switch, and one access switch.

Step 4, Confirm Legitimate Connectivity Still Works

Re-run the convergence test from Part 1, Step 4 (disconnect the same A1 uplink, ping, reconnect). Confirm the network still fails over correctly with all the new hardening in place.

Screenshot the successful ping recovery after the hardened re-test.

Step 5, Prove BPDU Guard and BPDU Filter Actually Work, and Compare Them

Add one more switch to your topology (representing a rogue device). You'll connect it to each protected port in turn and compare what happens.

5a. Test A1 (BPDU Guard)

Connect the rogue switch to A1's Fa0/1, the same port PC1 is on. Since a switch sends BPDUs and PC1 never did, this simulates exactly the kind of unauthorised device BPDU Guard is meant to catch.

```
show interfaces status
show interfaces fastEthernet 0/1
```

A1's Fa0/1 should now show as err-disabled. BPDU Guard detected the incoming BPDU and shut the port down automatically.

Screenshot the err-disabled port status on A1.

Remove the rogue switch and re-enable the port (shutdown / no shutdown) to restore PC1's connectivity before moving on.

5b. Test A2 (BPDU Filter)

Now connect the same rogue switch to A2's Fa0/1 instead.

show interfaces status

show interfaces fastEthernet 0/1

Unlike A1, A2's Fa0/1 should still show as connected/up, not err-disabled. BPDU Filter doesn't shut the port down; it silently discards the rogue switch's BPDUs and carries on. The rogue device still can't influence your STP topology, but nothing was logged or flagged to tell you it was ever there.

Screenshot A2's port status showing it stayed up. Then remove the rogue switch before finishing.

Step 6, Reflect on the Comparison

In one or two sentences, note the practical difference you just observed between A1 and A2's behaviour when the rogue switch was connected, in your own words, before moving to the final reflection below.

Reflection

Answer in a short paragraph (4 to 6 sentences): comparing your Part 1 baseline (PVST+) convergence time to your Rapid PVST+ result, and then to what Step 5 demonstrated about BPDU Guard versus BPDU Filter, which of these observations do you think matters more for a real network's security posture, and which matters more for its day-to-day reliability? Are these the same thing? If you were choosing between BPDU Guard and BPDU Filter for a real corporate access port, which would you pick, and why?